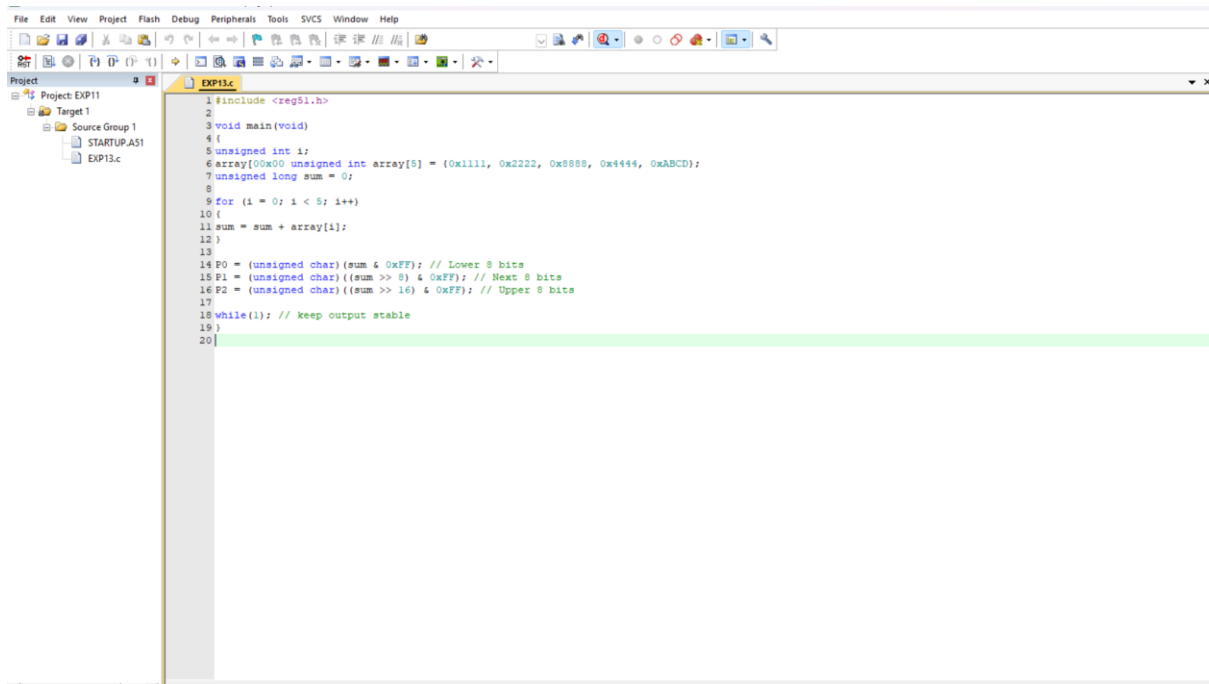


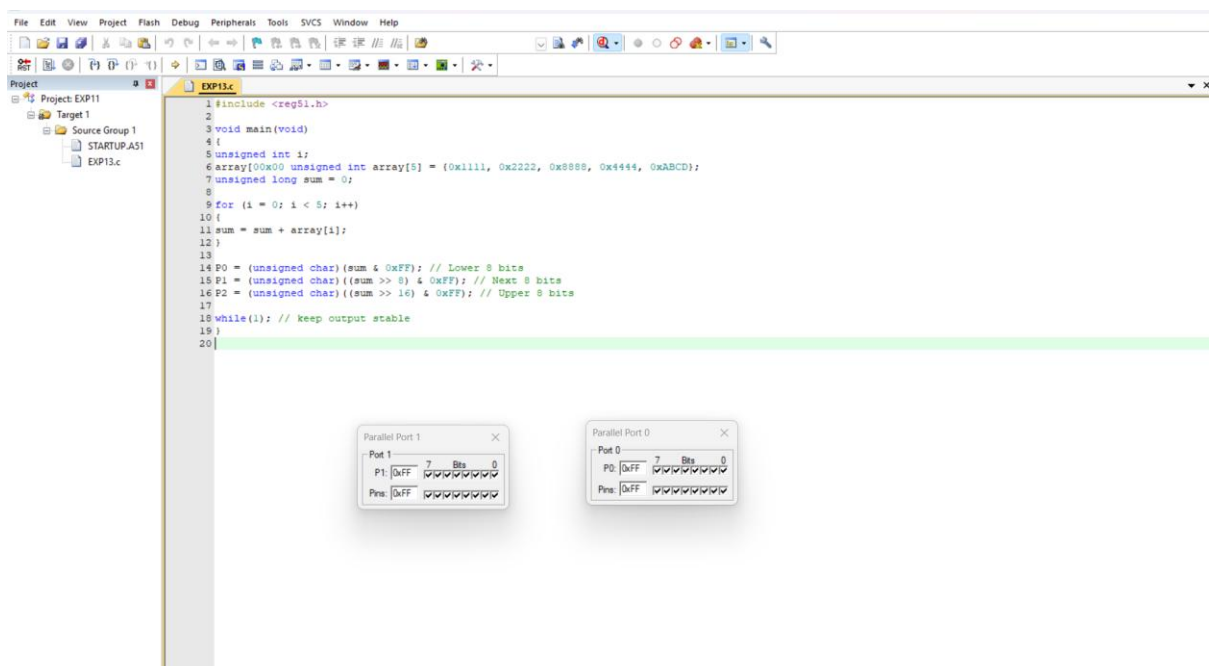
ECA1411-EMBEDDED SYSTEM

LAB MANUAL

Exp No:13- EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM



```
1#include <reg51.h>
2
3void main(void)
4{
5    unsigned int i;
6    array[0x00 unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
7    unsigned long sum = 0;
8
9    for (i = 0; i < 5; i++)
10    {
11        sum = sum + array[i];
12    }
13
14    P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
15    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
16    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
17
18    while(1); // keep output stable
19
20}
```



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```

Parallel Port 1

Port 1	7	Bits	0
P1	0xFF	✓✓✓✓✓✓✓✓	0
Pins	0xFF	✓✓✓✓✓✓✓✓	

Parallel Port 0

Port 0	7	Bits	0
P0	0xFF	✓✓✓✓✓✓✓✓	0
Pins	0xFF	✓✓✓✓✓✓✓✓	